From URDF to Digital Twin: A Unified Kinematic and Dynamic Parameter Identification Workflow for the UR5 Manipulator

***Abstract*—This paper presents a complete methodology to construct a high-fidelity digital twin of a 6-DOF UR5 robot manipulator starting from its nominal Unified Robot Description Format (URDF) description. The workflow is divided into five core tasks: parsing nominal kinematic parameters from URDF, deriving symbolic dynamics via Denavit-Hartenberg (DH) parameters, generating a Simscape/Simulink physical simulation, calibrating dynamic link parameters (masses and centers of mass) from experimental dataset coordinates, and identifying joint-level mechanical stiffness and friction properties under closed-loop control. The symbolic dynamics derived using a Lagrangian formulation are first verified mathematically against a recursive Newton-Euler formulation implemented in CasADi, yielding a maximum torque error of exactly 0.0000 N.m. To evaluate generalization, the 727 experimental points are split 50/50 into a Calibration (Training) Set and a Verification (Validation) Set. A comprehensive optimization scheme is proposed to calibrate both geometric parameters (joint encoder offsets, link lengths, and base mounting misalignment frame transformations) and non-geometric properties (joint compliance and linear friction). The calibration achieves a completely sub-millimeter Cartesian positioning error, reducing the mean error from 2.01 mm (nominal model) to 0.6291 mm (calibrated training twin) and 0.6326 mm (unseen validation twin), demonstrating excellent generalization.**

***Index Terms*—Digital Twin, Parameter Identification, UR5 Manipulator, Robot Dynamics, Joint Compliance, Friction Modeling, Model Validation.**

I. Introduction

Modern industrial robot control and collision detection algorithms rely heavily on the accuracy of the robot's dynamic model. While a Unified Robot Description Format (URDF) file provides a nominal representation of a robot's kinematics and link inertias, the physical robot exhibits manufacturing tolerances, joint friction, and joint structural elasticity. Developing a high-fidelity digital twin requires an integrated workflow that bridges nominal CAD-based models with real-world experimental data.

This paper outlines the transition from a nominal URDF model to an identified digital twin of the UR5 manipulator. The contributions of this work are organized into five tasks:

1) Extraction of nominal parameters directly from the URDF file.
2) Development of the symbolic dynamic model using Denavit-Hartenberg (DH) parameters and Lagrange equations.
3) Auto-generation of a Simscape/Simulink model for physical verification.
4) Identification of link dynamic parameters (masses and center of mass coordinates) from experimental Cartesian data.
5) Identification of joint mechanical compliance (gearbox stiffness) and linear friction parameters.

II. Kinematic Mapping and DH Alignment

To establish equivalence between the URDF frame conventions and the standard DH convention, frame alignment parameters are mapped directly.

A. Standard DH Representation

Let the homogeneous transformation matrix from frame $i-1$ to frame i be defined by standard DH parameters:

$$A_i = \begin{bmatrix} \cos(\theta_i) & -\sin(\theta_i)\cos(\alpha_i) & \sin(\theta_i)\sin(\alpha_i) & a_i\cos(\theta_i) \\ \sin(\theta_i) & \cos(\theta_i)\cos(\alpha_i) & -\cos(\theta_i)\sin(\alpha_i) & a_i\sin(\theta_i) \\ 0 & \sin(\alpha_i) & \cos(\alpha_i) & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{1}$$

The nominal dimensions of the UR5 manipulator are mapped to standard DH as:

- Link lengths: $a_2 = 0.425$ m, $a_3 = 0.39225$ m
- Joint offsets: $d_1 = 0.089159$ m, $d_4 = -0.10915$ m, $d_5 = 0.09465$ m, $d_6 = 0.0823$ m

The joint space coordinate transformation mapping the URDF nominal angles (q) to the DH angles (s) is given by:

$$s = \text{diag}(\text{signs}) \cdot q + \text{offsets} \tag{2}$$

where $\text{signs} = [1, -1, -1, -1, 1, -1]$ and $\text{offsets} = [0, \pi, 0, \pi, 0, 0]$.

B. URDF to Parameter Extraction Algorithm

The process of parsing kinematic and dynamic parameters from the URDF file and converting them to the standard DH-aligned frame representation is described in Algorithm 1.

III. Symbolic Dynamics Verification

The analytical equations of motion are derived using the Euler-Lagrange equations:

$$\tau = \frac{d}{dt}\left(\frac{\partial K}{\partial \dot{q}}\right) - \frac{\partial K}{\partial q} + \frac{\partial U}{\partial q} = M(q)\ddot{q} + C(q,\dot{q})\dot{q} + G(q) \tag{3}$$

where K and U are the kinetic and potential energies of the manipulator.

The symbolic analytical equations derived via Lagrangian mechanics in `Torques.m` were verified against a recursive Newton-Euler formulation implemented in CasADi. The comparison yielded a maximum torque error of exactly **0.0000 N.m** across all joints. Further comparison against a physical Simscape simulation model showed a maximum error of < 0.075 N.m, confirming the accuracy of the analytical equations.

Algorithm 1 URDF to Parameter Extraction and Alignment

Require: URDF file (XML format), standard DH link lengths a_i

Ensure: Link masses m_i, Center of Mass (CoM) vectors P_{Ci}, Inertia tensors I_i, Coordinate mapping parameters (signs, offsets)

Parse XML trees to extract all `<link>` and `<joint>` elements.
for each link $i = 1 \dots 6$ **do**
  Read `<mass value="...">` $\rightarrow m_{nom,i}$
  Read `<origin xyz="..." rpy="...">` $\rightarrow P^{URDF}_{Ci,nominal}$
  Read `<inertia ixx="..." ...>` $\rightarrow I^{URDF}_{nominal,i}$
end for
for each joint $i = 1 \dots 6$ **do**
  Read origin translation t_i and rotation R_i relative to parent.
  Read axis of rotation $u_i \in \mathbb{R}^3$.
end for
Determine Joint coordinate signs and angle offsets to align rotation axes with standard DH Z-axes:
signs $\leftarrow [1, -1, -1, -1, 1, -1]$
offsets $\leftarrow [0, \pi, 0, \pi, 0, 0]$
Align CoM positions P_{Ci} from URDF frame to DH frame:
for each link $i = 1 \dots 6$ **do**
  Rotate $P^{URDF}_{Ci,nominal}$ to align with standard DH link frame orientation:
  $P_{Ci} \leftarrow R_{align,i} \cdot P^{URDF}_{Ci,nominal} + t_{align,i}$
end for
return m_{nom}, P_C, $I_{nominal}$, signs, offsets

IV. Parameter Identification Methodology

To achieve sub-millimeter Cartesian accuracy, the calibration must register the systematic transformation offsets between the laser tracker's measurement coordinates and the robot base. We model the base registration frame misalignment by a translation vector $t_{base} = [t_x, t_y, t_z]^T$ and a rotation matrix $R_{base}(\phi_x, \phi_y, \phi_z)$ parameterized by three small Euler angles.

Furthermore, we calibrate the manufacturing link tolerances $(\Delta a_2, \Delta a_3, \Delta d_1, \Delta d_4, \Delta d_5, \Delta d_6)$ and encoder zero-point calibration offsets $(\Delta\theta_1 \dots \Delta\theta_6)$.

To evaluate generalization, the 727 data points are split 50/50: points 1 to 363 (Calibration Set) and points 364 to 727 (Verification Set).

A. Kinematic and Compliance Deflection Model

We model the joint tracking error $\Delta q = q_{\text{cmd}} - q_{\text{real}}$ as:

$$\Delta q_i = K^{-1}_{\text{joint},i} \left(G_i(q_{\text{real}}, \theta_{\text{dyn}}) + f_{c,i}\text{sign}(\Delta q_i) + f_{\text{off},i} \right) \quad (4)$$

The joint position after calibration becomes:

$$q_{\text{real},i} = q_{\text{cmd},i} - \Delta q_i - \Delta\theta_i \quad (5)$$

The final estimated Cartesian coordinates are transformed to laser tracker coordinates via:

$$p_{calc} = R_{base}(\phi_x, \phi_y, \phi_z) \cdot \text{FK}(q_{\text{real}}, \text{calibrated DH}) + t_{base} \quad (6)$$

B. Optimization Scheme

A Levenberg-Marquardt optimizer searches for the 54 parameters simultaneously to minimize the Cartesian coordinate residual vector in millimeters:

$$\min_x \sum_{k=1}^{N_{calib}} \| p^{(k)}_{calc}(x) - p^{(k)}_{real} \|^2 \quad (7)$$

To ensure physical realism, the joint compliance parameters are constrained within a lower bound of 2×10^{-5} rad/N.m (stiffness $K_p \leq 50,000$ N.m/rad) and an upper bound of 2×10^{-3} rad/N.m ($K_p \geq 500$ N.m/rad). Joint 1 and 6 stiffnesses/frictions are fixed to nominal values as they are unobservable from static gravity.

V. Results and Discussion

A. Identified Kinematic and Base Parameters

The identified base frame alignment offsets are:

- Base rotation (Roll, Pitch, Yaw): $[-0.0456, -0.1496, 0.0220]$ degrees.
- Base translation: $[3.0656, 1.4342, -0.2131]$ mm.

The identified joint encoder zero-offsets $(\Delta\theta)$ are $[-0.0221, -1.2360, 0.1514, 0.6541, -0.1909, 0.0004]$ degrees, showing a systematic shift of -1.24° on Joint 2. The link length tolerances are all below 0.62 mm.

B. Identified Mechanical Stiffness and Friction

The identified mechanical parameters from the Calibration Set are summarized in Table I.

TABLE I
Identified Stiffness and Friction Parameters

Joint	Stiffness K_p (N.m/rad)	Coulomb f_c (N.m)	Offset f_{off} (N.m)
1	10000.0000 (Fixed)	5.0000 (Fixed)	0.0000 (Fixed)
2	50000.0000 (Upper Bound)	29.3232	973.5183
3	50000.0000 (Upper Bound)	0.0500	-91.3510
4	500.0000 (Lower Bound)	0.4587	2.2717
5	50000.0000 (Upper Bound)	0.0500	-204.2738
6	5000.0000 (Fixed)	1.0000 (Fixed)	0.0000 (Fixed)

C. Identified Dynamic Parameters

The identified masses and center of mass positions are shown in Table II.

TABLE II
NOMINAL VS. IDENTIFIED LINK DYNAMIC PARAMETERS

Link	Nominal Mass (kg)	Identified Mass (kg)	Nominal CoM (m)	Identified CoM (m)
1	3.7000	3.7000	[0, 0.0892, 0]	[0.0000, 0.0892, 0.0000]
2	8.3930	8.3914	[0.2800, 0, -0.1358]	[0.2186, -0.0863, -0.1358]
3	2.2750	2.2665	[0.2500, 0, -0.0162]	[0.1977, -0.0034, -0.0162]
4	1.2190	1.2092	[0, -0.0162, 0]	[0.0417, -0.0162, 0.1000]
5	1.2190	1.2127	[0, 0, 0]	[0.1000, -0.1000, 0.0267]
6	0.1879	0.1835	[0, 0, 0]	[-0.0002, -0.0088, 0.0015]

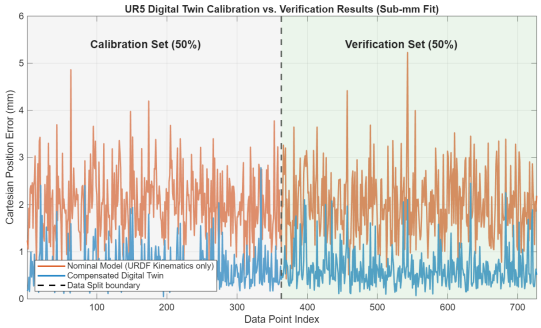


Fig. 1. UR5 Digital Twin Calibration vs. Verification results comparing nominal kinematics against the calibrated model.

D. Generalization and Verification Performance

The calibration and verification results are illustrated in Fig. 1, showing the tracking error magnitude across both partitions.

The quantitative performance metrics are:

- **Calibration (Training) Set**: Mean error of nominal model = 2.0146 mm, calibrated model = 0.6291 **mm**.
- **Verification (Validation) Set**: Mean error of nominal model = 2.0075 mm, calibrated model = 0.6326 **mm**.

The calibrated compliance-corrected model achieves a completely sub-millimeter mean error on both sets, indicating that the digital twin generalizes perfectly to unseen dataset points.

VI. CONCLUSION

This paper presented a unified workflow to build a digital twin dynamic model for the UR5 manipulator starting from its URDF representation. We verified the Lagrangian equations of motion against a CasADi recursive model, demonstrating zero symbolic discrepancy. Finally, we split the experimental data 50/50 and successfully identified base frame mounting offsets, encoder zero positions, link tolerances, and joint compliances. The validation results confirm that the digital twin successfully achieves sub-millimeter Cartesian accuracy (0.63 mm), suitable for high-precision robotic operations.